

Battery Sizing for the Delivery Cart Robot

1. Design basis and system architecture

The robot uses a **12 V Li-ion main battery pack with internal BMS**. This battery powers the full electrical system **except the Raspberry Pi**, which is powered separately through a power bank. Therefore, the Raspberry Pi and its traffic light module are **excluded** from the main battery sizing.

The main battery is responsible for supplying:

- four 12 V DC drive motors
- two H-bridges
- one stepper motor for the scissor-lift mechanism
- one DRV8825 stepper driver
- six IR sensors
- four motor encoders
- two level shifters
- one ESP controller
- one MPU IMU
- one buzzer
- the 5 V rails through buck converters

The robot also includes:

- an internal battery BMS
- a reverse polarity safety circuit using MOSFET and resistors
- a fuse for overcurrent protection
- an emergency stop button
- charging provision

This means the battery sizing must satisfy both the **energy requirement** and the **current capability requirement**, while also respecting the system safety architecture.

2. Main assumptions used in the sizing

Because exact datasheet currents are not yet available for all devices, the following assumptions are used.

Battery assumptions

Parameter	Value
Battery chemistry	Li-ion
Battery system voltage used in calculations	12 V
Usable depth of discharge	80%
Overall design safety factor	1.3
Buck converter efficiency	90%

Actuator assumptions

Parameter	Value
Number of DC drive motors	4
Average current per DC motor	0.3 to 0.6 A
Stepper motor current	2.0 A
Stepper operating state during mission	Always energized, including holding

The stepper motor is treated as a **continuous load**. Therefore:

$$I_{stepper} = 2.0 \text{ A}$$

3. Load table and operating voltages

3.1 12 V loads

Component	Quantity	Operating Voltage	Assumed Current	Notes
DC drive motor	4	12 V	0.3 to 0.6 A each	average running current
Stepper motor	1	12 V through DRV8825	2.0 A total	continuously energized for holding and lifting
H-bridge boards	2	12 V motor side	included with motor load	board losses neglected in first sizing
DRV8825 driver (VMOT side)	1	12 V motor supply	included with stepper load	motor power path
Reverse polarity circuit	1	12 V line	negligible	safety element
Fuse	1	12 V line	negligible	protection element
E-stop button	1	actuator line	negligible	control/safety element

3.2 5 V loads supplied through buck converter(s)

Component	Quantity	Operating Voltage	Assumed Current Each	Total Current
IR sensors	6	5 V	0.020 A	0.120 A
Motor encoders	4	5 V	0.020 A	0.080 A
Level shifters	2	5 V	0.005 A	0.010 A
ESP board	1	5 V	0.100 A	0.100 A
MPU IMU	1	5 V	0.005 A	0.005 A
Buzzer	1	5 V	0.010 A average	0.010 A
DRV8825 VDD input	1	5 V	0.010 A	0.010 A
Driver/control overhead	1 lumped	5 V	0.040 A	0.040 A

So the total 5 V load current is:

$$I_{5V,total} = 0.120 + 0.080 + 0.010 + 0.100 + 0.005 + 0.010 + 0.010 + 0.040$$
$$I_{5V,total} = 0.375 \text{ A}$$

4. Buck converter check

Your buck converter provides:

- **5 V**
- **3 A output**

The required 5 V load current is:

$$0.375 \text{ A}$$

So one 5 V, 3 A buck converter is already sufficient, because:

$$\frac{0.375}{3} \times 100 = 12.5\%$$

This means even a single buck would be loaded to only **12.5% of its rated current**. Since you already have two buck converters, this gives very good margin and flexibility.

A practical distribution consistent with your actual wiring is:

- **Buck 1:** ESP + MPU + IR sensors + encoders + level shifters + buzzer + general control overhead
- **Buck 2:** DRV8825 VDD input only

The stepper motor itself is **not** powered through the 5 V buck. It remains on the **12 V actuator rail** through the DRV8825 motor supply path.

5. Converting the 5 V electronics load into equivalent battery current

First, the total power required by the 5 V electronics is:

$$P_{5V} = V \times I$$
$$P_{5V} = 5 \times 0.375 = 1.875 \text{ W}$$

Now convert this to equivalent current drawn from the 12 V battery, using 90% buck efficiency:

$$I_{battery,aux} = \frac{P_{5V}}{V_{battery} \times \eta}$$
$$I_{battery,aux} = \frac{1.875}{12 \times 0.9}$$
$$I_{battery,aux} = 0.174 \text{ A}$$

So the always-on electronics contribute approximately:

$$I_{battery,aux} \approx 0.174 \text{ A}$$

For clarity, the two-buck split can also be checked separately.

Buck 1 load

Buck 1 supplies:

- IR sensors = 0.120 A
- encoders = 0.080 A
- level shifters = 0.010 A
- ESP = 0.100 A
- MPU = 0.005 A
- buzzer = 0.010 A
- general control overhead = 0.040 A

So:

$$I_{Buck1} = 0.120 + 0.080 + 0.010 + 0.100 + 0.005 + 0.010 + 0.040$$
$$I_{Buck1} = 0.365 \text{ A}$$

Its power is:

$$P_{Buck1} = 5 \times 0.365 = 1.825 \text{ W}$$

Equivalent battery current:

$$I_{battery,Buck1} = \frac{1.825}{12 \times 0.9} = 0.169 \text{ A}$$

Buck 2 load

Buck 2 supplies only the DRV8825 VDD input:

$$I_{Buck2} = 0.010 \text{ A}$$

Its power is:

$$P_{Buck2} = 5 \times 0.010 = 0.050 \text{ W}$$

Equivalent battery current:

$$I_{battery,Buck2} = \frac{0.050}{12 \times 0.9} = 0.0046 \text{ A}$$

Therefore:

$$\begin{aligned} I_{battery,aux} &= I_{battery,Buck1} + I_{battery,Buck2} \\ I_{battery,aux} &= 0.169 + 0.0046 \\ I_{battery,aux} &\approx 0.174 \text{ A} \end{aligned}$$

6. Mission profile model

Since the robot is expected to move continuously while sensing obstacles, the sizing can be simplified into two operating conditions.

Base condition

This is the dominant condition during the mission. In this state:

- four DC motors are running
- IR sensors are active
- encoders are active
- ESP is active
- MPU is active
- supporting electronics are active
- DRV8825 VDD logic input is powered from Buck 2
- the stepper motor is energized and holding position

Lift condition

This occurs when the scissor mechanism is commanded to raise. In this state:

- all base-condition loads are still active
- the stepper motor remains energized and performs motion

Because the stepper is assumed to remain energized all the time, its current is treated as a **continuous contribution** throughout the mission rather than duty-cycled.

Thus, the stepper's average contribution becomes:

$$I_{stepper,avg} = 2.0 \text{ A}$$

7. Average battery current calculation

7.1 Drive motor contribution

Each motor is assumed to draw:

$$0.3 \text{ to } 0.6 \text{ A}$$

With four motors:

$$I_{drive} = 4 \times (0.3 \text{ to } 0.6)$$

$$I_{drive} = 1.2 \text{ to } 2.4 \text{ A}$$

7.2 Total average battery current

Now add the battery-side auxiliary load and the stepper average contribution:

$$I_{avg,total} = I_{drive} + I_{battery,aux} + I_{stepper,avg}$$

$$I_{avg,total} = (1.2 \text{ to } 2.4) + 0.174 + 2.0$$

$$I_{avg,total} = 3.374 \text{ to } 4.574 \text{ A}$$

So the total average current required from the main battery is:

$$I_{avg,total} = 3.37 \text{ to } 4.57 \text{ A}$$

These updated current bounds are the basis for the rest of the sizing.

8. Battery capacity equation

Battery capacity is determined from:

$$C_{required} = \frac{I_{avg,total} \times t \times SF}{DoD}$$

Where:

- $I_{avg,total}$ is the average battery current
- t is runtime in hours

- $SF = 1.3$
- $DoD = 0.8$

So:

$$C_{required} = I_{avg,total} \times t \times \frac{1.3}{0.8}$$

$$C_{required} = I_{avg,total} \times t \times 1.625$$

9. Capacity calculations for the requested runtime range

9.1 For 25 minutes runtime

$$t = \frac{25}{60} = 0.4167 \text{ h}$$

Lower-current case

$$C_{25,min} = 3.374 \times 0.4167 \times 1.625$$

$$C_{25,min} = 2.28 \text{ Ah}$$

Higher-current case

$$C_{25,max} = 4.574 \times 0.4167 \times 1.625$$

$$C_{25,max} = 3.10 \text{ Ah}$$

So for a 25-minute mission:

$C_{25} = 2.28 \text{ to } 3.10 \text{ Ah}$

9.2 For 30 minutes runtime

$$t = 0.5 \text{ h}$$

Lower-current case

$$C_{30,min} = 3.374 \times 0.5 \times 1.625$$

$$C_{30,min} = 2.74 \text{ Ah}$$

Higher-current case

$$C_{30,max} = 4.574 \times 0.5 \times 1.625$$

$$C_{30,max} = 3.72 \text{ Ah}$$

So for a 30-minute mission:

$$C_{30} = 2.74 \text{ to } 3.72 \text{ Ah}$$

9.3 For 45 minutes runtime

$$t = 0.75 \text{ h}$$

Lower-current case

$$C_{45,min} = 3.374 \times 0.75 \times 1.625$$

$$C_{45,min} = 4.11 \text{ Ah}$$

Higher-current case

$$C_{45,max} = 4.574 \times 0.75 \times 1.625$$

$$C_{45,max} = 5.57 \text{ Ah}$$

So for a 45-minute mission:

$$C_{45} = 4.11 \text{ to } 5.57 \text{ Ah}$$

9.4 For 60 minutes runtime

$$t = 1.0 \text{ h}$$

Lower-current case

$$C_{60,min} = 3.374 \times 1.0 \times 1.625$$

$$C_{60,min} = 5.48 \text{ Ah}$$

Higher-current case

$$C_{60,max} = 4.574 \times 1.0 \times 1.625$$

$$C_{60,max} = 7.43 \text{ Ah}$$

So for a 60-minute mission:

$C_{60} = 5.48 \text{ to } 7.43 \text{ Ah}$

All of these capacity values come directly from the updated current range and runtime equation.

10. Energy values in watt-hours

For completeness, the corresponding nominal battery energy can be estimated as:

$$E = V \times C$$

Using 12 V:

25 minutes

$$E_{25} = 12 \times (2.28 \text{ to } 3.10)$$

$$E_{25} = 27.4 \text{ to } 37.2 \text{ Wh}$$

45 minutes

$$E_{45} = 12 \times (4.11 \text{ to } 5.57)$$

$$E_{45} = 49.3 \text{ to } 66.9 \text{ Wh}$$

60 minutes

$$E_{60} = 12 \times (5.48 \text{ to } 7.43)$$

$$E_{60} = 65.8 \text{ to } 89.2 \text{ Wh}$$

These energy values reflect the regenerated battery-capacity range.

11. Interpretation of the results

The calculated battery capacity values are now higher than the very low-stepper case because:

- the Raspberry Pi is still powered from a separate power bank
- the low-power electronics still consume very little current
- the four drive motors still add **1.2 to 2.4 A**
- the stepper motor is now treated as a **continuous 2.0 A load**
- therefore, the final battery size depends mainly on the drive motors plus the continuously energized stepper

However, in real mobile robots, especially those using **four mecanum wheels**, current can rise noticeably due to:

- starting current
- turning losses
- wheel slip
- floor friction
- payload variation
- obstacle avoidance maneuvers
- battery aging
- voltage sag under load

Therefore, although the mathematical requirement for a 60-minute mission is approximately **5.48 to 7.43 Ah**, it is not good engineering practice to select a battery exactly equal to this value.

12. Final battery selection

Minimum technically acceptable choice

For a one-hour mission based on the present assumptions, the battery should be at least:

12 V Li-ion, 8 Ah

This covers the upper calculated result of 7.43 Ah with a small additional margin.

Practical recommended choice

For a safer and more robust design, the better selection is:

12 V Li-ion, 8 to 10 Ah

Recommended final choice for the report

If you want one clean recommended number to defend in the document, I suggest:

12 V	L
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This is a very reasonable selection because it gives:

- margin beyond the calculated one-hour requirement
- better tolerance to motor current uncertainty
- better tolerance to mecanum drive losses
- more stable voltage under transients
- room for future additions

13. Peak current and BMS selection

Battery sizing is not only about Ah. The battery must also support short-duration current peaks.

Since exact motor stall current is not available, a conservative engineering estimate is to assume each motor may momentarily draw around **3 times its average current** during startup or aggressive movement.

Thus, the motor peak range becomes:

$$I_{motor,peak} = 4 \times (0.9 \text{ to } 1.8)$$

$$I_{motor,peak} = 3.6 \text{ to } 7.2 \text{ A}$$

Now add:

- stepper motor continuous current = **2.0 A**
- auxiliary battery-side load = **0.174 A**

So the approximate system peak becomes:

$$I_{peak,total} = 3.6 \text{ to } 7.2 + 2.0 + 0.174$$

$$I_{peak,total} = 5.77 \text{ to } 9.37 \text{ A}$$

Because this is still based on assumed motor behavior, the battery/BMS should be selected with clear margin.

Recommended BMS rating

BMS c	ontinuous c
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and preferably:

BMS c	ontinuous c
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The regenerated peak-current range supports this recommendation.

14. Fuse selection

The fuse must be high enough to survive normal current peaks, but low enough to protect wiring and electronics under fault conditions.

Since the estimated peak current can approach roughly **9 to 10 A**, a reasonable preliminary fuse selection is:

15 A	f
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This is a good practical choice because it is:

- above the expected average operating current
- above likely transient current peaks
- still low enough to provide meaningful fault protection

If later testing shows much higher startup current, then the fuse may be adjusted accordingly.

16. Short final design statement for the report

The robot is powered by a 12 V Li-ion main battery pack with internal BMS, while the Raspberry Pi is supplied separately by a power bank and is therefore excluded from the main battery sizing. The main battery powers the four 12 V DC drive motors, the stepper-based scissor-lift mechanism through a DRV8825 driver, the ESP controller, the IR sensors, encoders, level shifters, IMU, buzzer, and the 5 V auxiliary rail through buck conversion. The DRV8825 motor supply remains on the 12 V actuator rail, while a separate 5 V buck converter is connected only to the DRV8825 VDD input. Based on an assumed average current of 0.3 to 0.6 A per drive motor, a continuously energized stepper current of 2.0 A, and a total 5 V auxiliary current of 0.375 A, the equivalent average main-battery current is estimated to be 3.37 to 4.57 A. After applying a design safety factor of 1.3 and limiting usable depth of discharge to 80%, the required battery capacity is approximately 2.28 to 3.10 Ah for 25 minutes, 4.11 to 5.57 Ah for 45 minutes, and 5.48 to 7.43 Ah for 60 minutes of operation. To ensure reliable real-world performance and to account for uncertainty in actual motor current, a practical battery selection of 12 V, 10 Ah Li-ion is recommended. A BMS rating of at least 15 A, preferably 20 A, together with a 15 A fuse, is considered appropriate for the current design stage.

17. Final selected values

Item	Selected / Recommended Value
Main battery chemistry	Li-ion
Main battery voltage	12 V
Minimum acceptable battery size	8 Ah
Recommended battery size	10 Ah
Buck converter rating	5 V, 3 A
Required total 5 V load current	0.375 A
Buck 1 output current	0.365 A
Buck 2 output current	0.010 A
Recommended BMS current rating	15 to 20 A
Recommended fuse	15 A
Stepper motor supply path	12 V through DRV8825 VMOT
Stepper driver VDD supply path	5 V through Buck 2
Raspberry Pi supply	separate power bank